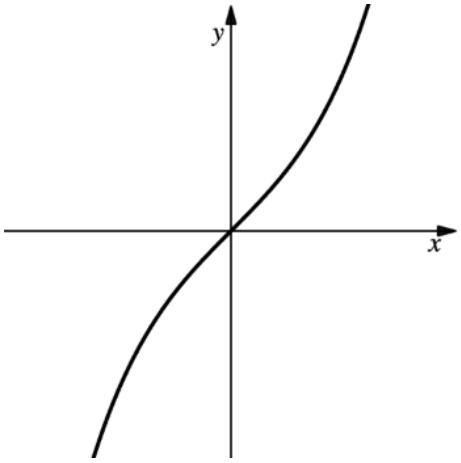
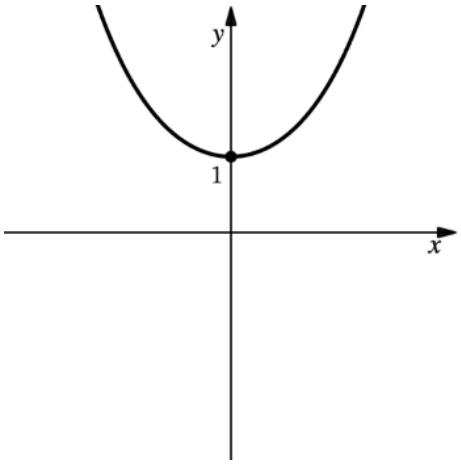
1) a)



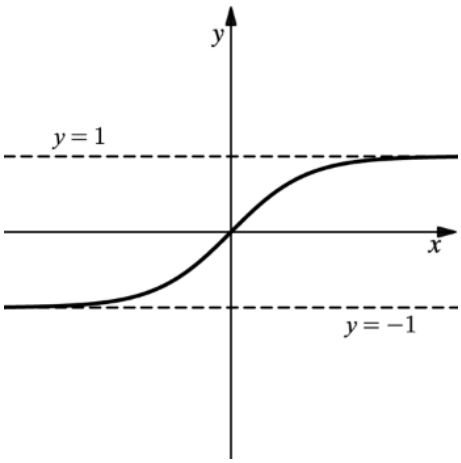
$y = \sinh x$ is one-to-one

1) b)



$y = \cosh x$ is many-to-one

1) c)



$y = \tanh x$ is one-to-one

2) a)

$y = \sinh x$

Domain: $\{x: x \in \mathbb{R}\}$

Range: $\{y: y \in \mathbb{R}\}$

2) b)

$y = \cosh x$

Domain: $\{x: x \in \mathbb{R}\}$

Range: $\{y: y \in \mathbb{R}, y \geq 1\}$

2) c)

$y = \tanh x$

Domain: $\{x: x \in \mathbb{R}\}$

Range: $\{y: y \in \mathbb{R}, -1 < y < 1\}$

3) a)

$y = \sinh x$

Range: $\{y: y \in \mathbb{R}, y < 0\}$

3) b)

$y = \cosh x$

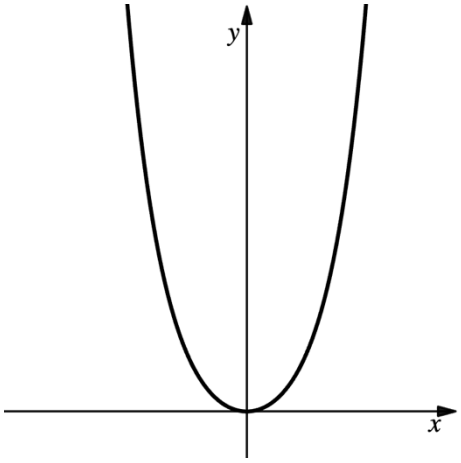
Range: $\{y: y \in \mathbb{R}, y \geq 1\}$

3) c)

$y = \tanh x$

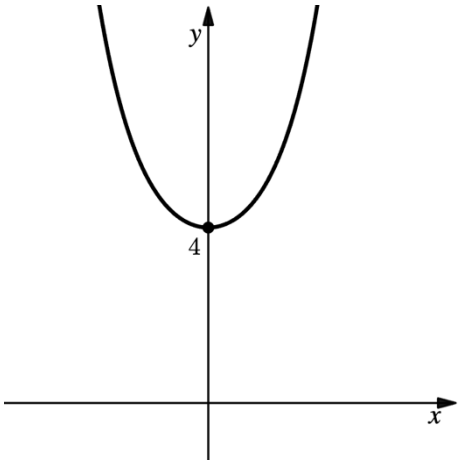
Range: $\{y: y \in \mathbb{R}, -1 < y < 0\}$

4) a)



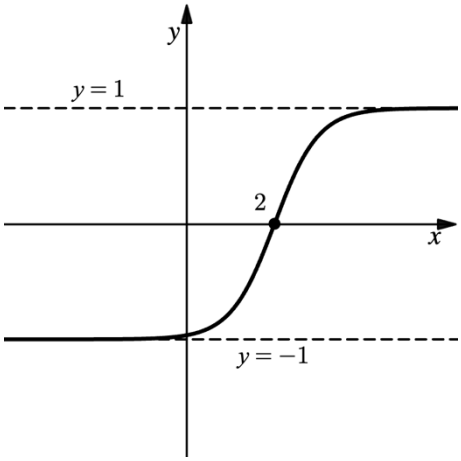
Range: $\{f(x): f(x) \in \mathbb{R}, f(x) \geq 0\}$

4) b)



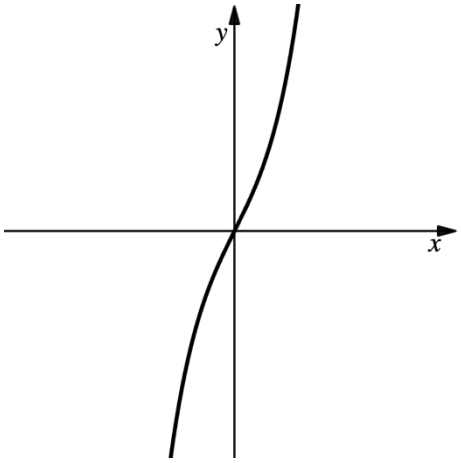
Range: $\{y: y \in \mathbb{R}, y \geq 4\}$

4) c)



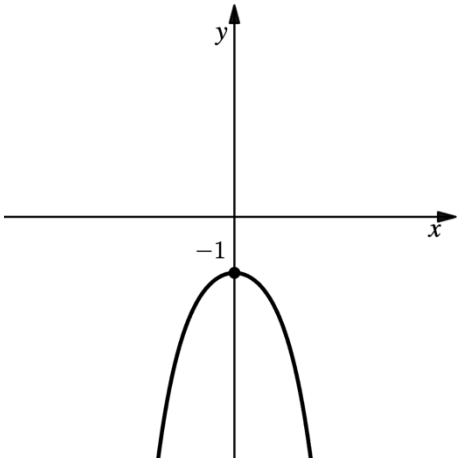
Range: $\{g(x): g(x) \in \mathbb{R}, -1 < g(x) < 1\}$

4) d)



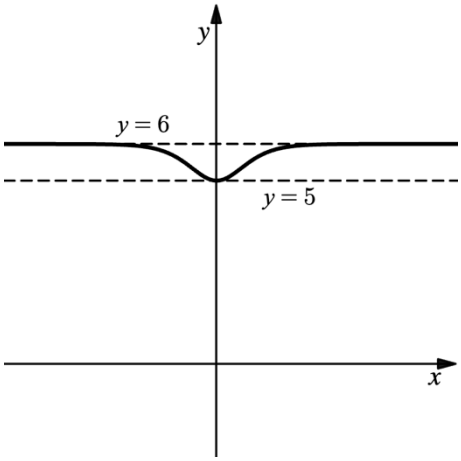
Range: $\{y: y \in \mathbb{R}\}$

4) e)



Range: $\{\mathrm{h}(x): \mathrm{h}(x) \in \mathbb{R}, \mathrm{h}(x) \leq -1\}$

4) f)



Range: $\{y: y \in \mathbb{R}, 5 \leq y < 6\}$

5) a)

$2\sinh x + 4\cosh x$

$\equiv 2 \times \frac{e^x - e^{-x}}{2} + 4 \times \frac{e^x + e^{-x}}{2}$

$\equiv e^x - e^{-x} + 2e^x + 2e^{-x}$

$\equiv 3e^x + e^{-x}$

5) b)

$3e^x > 0$ for all $x \in \mathbb{R}$

So $3e^x + e^{-x} > e^{-x}$

$\therefore 2\sinh x + 4\cosh x > e^{-x}$

6)

$4\sinh^2 x - e^{-2x}$

$\equiv 4\left(\frac{e^x - e^{-x}}{2}\right)^2 - e^{-2x}$

$\equiv (e^x - e^{-x})^2 - e^{-2x}$

$\equiv e^{2x} - 2 + e^{-2x} - e^{-2x}$

$\equiv e^{2x} - 2$

$e^{2x} > 0$ for all $x \in \mathbb{R}$

So $e^{2x} - 2 > -2$

$\therefore 4\sinh^2 x - e^{-2x} > -2$

7) a)

$\sinh 2x \equiv 2\sinh x \cosh x$

7) b) i)

$\{y : y \in \mathbb{R}, y > 0\}$

7) b) ii)

$\{y : y \in \mathbb{R}, y > 1\}$

7) c)

$\sinh 2x = 2\sinh x \cosh x$

From b) i) we know that $\sinh x > 0$ and from b) ii) we know that $\cosh x > 1$

So $2\cosh x > 2$

Multiply both sides by $\sinh x$ (which maintains the inequality as $\sinh x > 0$):

$2\sinh x \cosh x > 2\sinh x$

Also we know that $2\sinh x > \sinh x$ as $\sinh x > 0$

$\therefore \sinh 2x = 2\sinh x \cosh x > 2\sinh x > \sinh x$

$\Rightarrow \sinh 2x > \sinh x$ for $x > 0$

8)

$\mathrm{f}(x) = \sinh x + 2\cosh x$

$\mathrm{f}'(x) = \cosh x + 2\sinh x$

Stationary points exist when $\mathrm{f}'(x) = 0$

$\Rightarrow \cosh x + 2\sinh x = 0 \Rightarrow 1 + 2\tanh x = 0 \Rightarrow \tanh x = -\frac{1}{2}$

$\Rightarrow x = \frac{1}{2}\ln\left(\frac{1+\left(-\frac{1}{2}\right)}{1-\left(-\frac{1}{2}\right)}\right) = \frac{1}{2}\ln\left(\frac{\frac{1}{2}}{\frac{3}{2}}\right) = \frac{1}{2}\ln\frac{1}{3}$

$\mathrm{f}\left(\frac{1}{2}\ln\frac{1}{3}\right) = \sinh\left(\frac{1}{2}\ln\frac{1}{3}\right) + 2\cosh\left(\frac{1}{2}\ln\frac{1}{3}\right)$

$= \frac{\mathrm{e}^{\frac{1}{2}\ln\frac{1}{3}} - \mathrm{e}^{-\frac{1}{2}\ln\frac{1}{3}}}{2} + 2 \times \frac{\mathrm{e}^{\frac{1}{2}\ln\frac{1}{3}} + \mathrm{e}^{-\frac{1}{2}\ln\frac{1}{3}}}{2}$

$= \frac{\mathrm{e}^{\ln\left(\frac{1}{\sqrt{3}}\right)} - \mathrm{e}^{\ln(\sqrt{3})}}{2} + \mathrm{e}^{\ln\left(\frac{1}{\sqrt{3}}\right)} + \mathrm{e}^{\ln(\sqrt{3})}$

$= \frac{3}{2}\mathrm{e}^{\ln\left(\frac{1}{\sqrt{3}}\right)} + \frac{1}{2}\mathrm{e}^{\ln(\sqrt{3})}$

$= \frac{3}{2\sqrt{3}} + \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} = \sqrt{3}$

Range is $\left\{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \geq \sqrt{3}\right\}$

9)

$\mathrm{f}(x) = 3\cosh x - \sinh x \cosh x$

$\mathrm{f}'(x) = 3\sinh x - \sinh^2 x - \cosh^2 x$

Stationary points exist when $\mathrm{f}'(x) = 0$

$\Rightarrow 3\sinh x - \sinh^2 x - \cosh^2 x = 0$

$\Rightarrow 3\sinh x - \sinh^2 x - (\sinh^2 x + 1) = 0$

$\Rightarrow 0 = 2\sinh^2 x - 3\sinh x + 1$

$\Rightarrow 0 = (2\sinh x - 1)(\sinh x - 1)$

$\sinh x = \frac{1}{2}$ or $\sinh x = 1$

$\sinh x = \frac{1}{2} \Rightarrow x = \ln\left(\frac{1}{2} + \sqrt{\left(\frac{1}{2}\right)^2 + 1}\right) = \ln\left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right)$

$\sinh x = 1 \Rightarrow x = \ln\left(1 + \sqrt{1^2 + 1}\right) = \ln\left(1 + \sqrt{2}\right)$

When $x = \ln\left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right)$:

$\mathrm{f}\left(\ln\left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right)\right) = 3\cosh\left(\ln\left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right)\right) - \sinh\left(\ln\left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right)\right)\cosh\left(\ln\left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right)\right)$

$= 3 \times \frac{\mathrm{e}^{\ln\left(\frac{1}{2}+\frac{\sqrt{5}}{2}\right)} + \mathrm{e}^{-\ln\left(\frac{1}{2}+\frac{\sqrt{5}}{2}\right)}}{2} - \frac{\mathrm{e}^{\ln\left(\frac{1}{2}+\frac{\sqrt{5}}{2}\right)} - \mathrm{e}^{-\ln\left(\frac{1}{2}+\frac{\sqrt{5}}{2}\right)}}{2} \times \frac{\mathrm{e}^{\ln\left(\frac{1}{2}+\frac{\sqrt{5}}{2}\right)} + \mathrm{e}^{-\ln\left(\frac{1}{2}+\frac{\sqrt{5}}{2}\right)}}{2}$

$= 3 \times \frac{\frac{1}{2}+\frac{\sqrt{5}}{2}+\frac{\sqrt{5}}{2}-\frac{1}{2}}{2} - \frac{\frac{1}{2}+\frac{\sqrt{5}}{2}-\left(\frac{\sqrt{5}}{2}-\frac{1}{2}\right)}{2} \times \frac{\frac{1}{2}+\frac{\sqrt{5}}{2}+\frac{\sqrt{5}}{2}-\frac{1}{2}}{2}$

$= \frac{3\sqrt{5}}{2} - \frac{1}{2} \times \frac{\sqrt{5}}{2} = \frac{3\sqrt{5}}{2} - \frac{\sqrt{5}}{4} = \frac{5\sqrt{5}}{4} \approx 2.795$

When $x = \ln\left(1 + \sqrt{2}\right)$:

$\mathrm{f}\left(\ln\left(1 + \sqrt{2}\right)\right) = 3\cosh\left(\ln\left(1 + \sqrt{2}\right)\right) - \sinh\left(\ln\left(1 + \sqrt{2}\right)\right)\cosh\left(\ln\left(1 + \sqrt{2}\right)\right)$

$= 3 \times \frac{\mathrm{e}^{\ln(1+\sqrt{2})} + \mathrm{e}^{-\ln(1+\sqrt{2})}}{2} - \frac{\mathrm{e}^{\ln(1+\sqrt{2})} - \mathrm{e}^{-\ln(1+\sqrt{2})}}{2} \times \frac{\mathrm{e}^{\ln(1+\sqrt{2})} + \mathrm{e}^{-\ln(1+\sqrt{2})}}{2}$

$= 3 \times \frac{1+\sqrt{2}+\sqrt{2}-1}{2} - \frac{1+\sqrt{2}-(\sqrt{2}-1)}{2} \times \frac{1+\sqrt{2}+\sqrt{2}-1}{2}$

$= 3\sqrt{2} - \sqrt{2} = 2\sqrt{2} \approx 2.828$

Range is $\mathrm{f}(x) \in \left(\frac{5\sqrt{5}}{4}, 2\sqrt{2}\right]$

10) a)

$$f(x) = \frac{e^{2x}+e^{-2x}}{(e^x+e^{-x})^2} - \tanh^2 x$$

$$\equiv \frac{e^{2x}+e^{-2x}}{(e^x+e^{-x})^2} - \frac{\sinh^2 x}{\cosh^2 x}$$

$$\equiv \frac{e^{2x}+e^{-2x}}{(e^x+e^{-x})^2} - \frac{\left(\frac{e^x-e^{-x}}{2}\right)^2}{\left(\frac{e^x+e^{-x}}{2}\right)^2}$$

$$\equiv \frac{e^{2x}+e^{-2x}}{(e^x+e^{-x})^2} - \frac{(e^x-e^{-x})^2}{(e^x+e^{-x})^2}$$

$$\equiv \frac{e^{2x}+e^{-2x}}{(e^x+e^{-x})^2} - \frac{e^{2x}-2+e^{-x}}{(e^x+e^{-x})^2}$$

$$\equiv \frac{e^{2x}+e^{-2x}-e^{2x}+2-e^{-x}}{(e^x+e^{-x})^2}$$

$$\equiv \frac{2}{(e^x+e^{-x})^2}$$

$$\equiv 2(e^x + e^{-x})^{-2}$$

10) b)

Stationary points exist when $f'(x) = 0$

$$f'(x) = -4(e^x - e^{-x})(e^x + e^{-x})^{-3}$$

$$\equiv -\frac{4(e^x-e^{-x})}{(e^x+e^{-x})^3}$$

$$f'(x) = 0 \Rightarrow -\frac{4(e^x-e^{-x})}{(e^x+e^{-x})^3} = 0$$

$$\Rightarrow e^x - e^{-x} = 0$$

$$\Rightarrow e^{2x} - 1 = 0$$

$$\Rightarrow e^{2x} = 1$$

$$\Rightarrow 2x = 0$$

$$\Rightarrow x = 0$$

$$f(0) = 2(e^0 + e^0)^{-2} = \frac{1}{2}$$

So the stationary point has coordinates $\left(0, \frac{1}{2}\right)$

10) c)

As $x \to +\infty$, $\frac{2}{(e^x+e^{-x})^2} \to 0$

As $x \to -\infty$, $\frac{2}{(e^x+e^{-x})^2} \to 0$

So the range of f is $\left\{f(x): f(x) \in \mathbb{R}, 0 < f(x) \leq \frac{1}{2}\right\}$